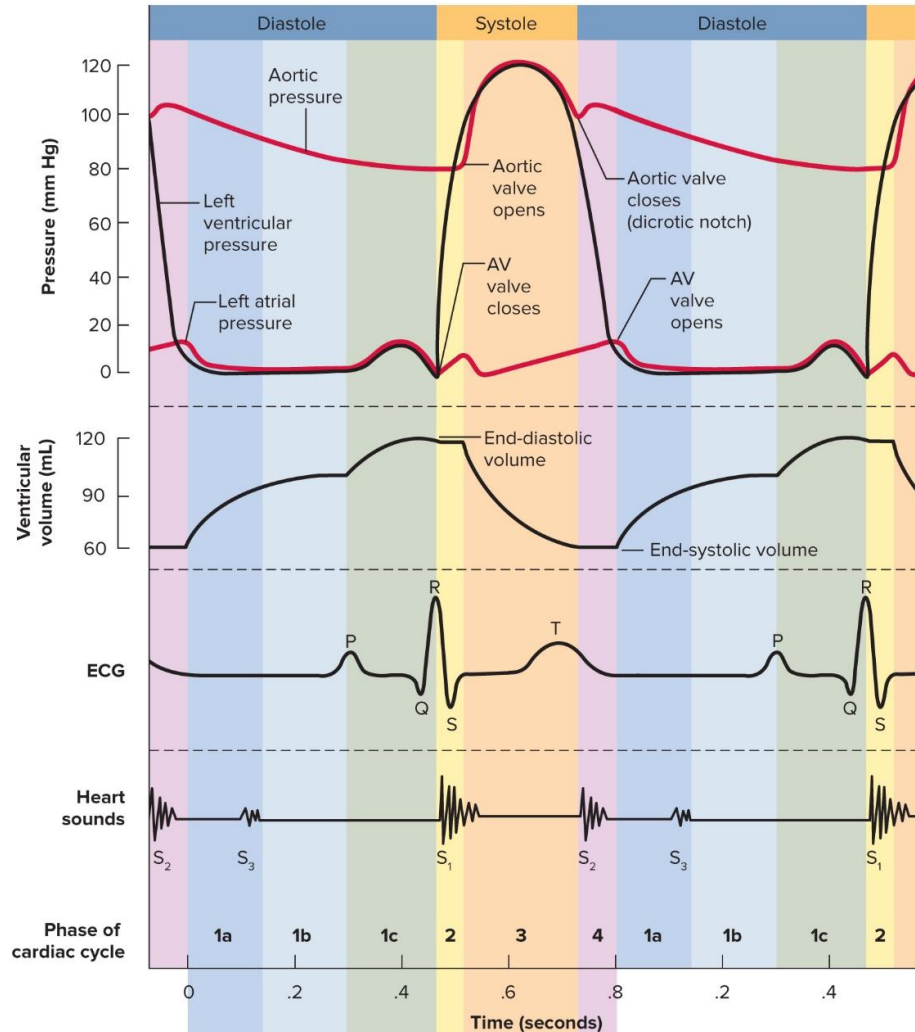


Exercise 16: Blood Pressure & Lymphatic



Blood Pressure



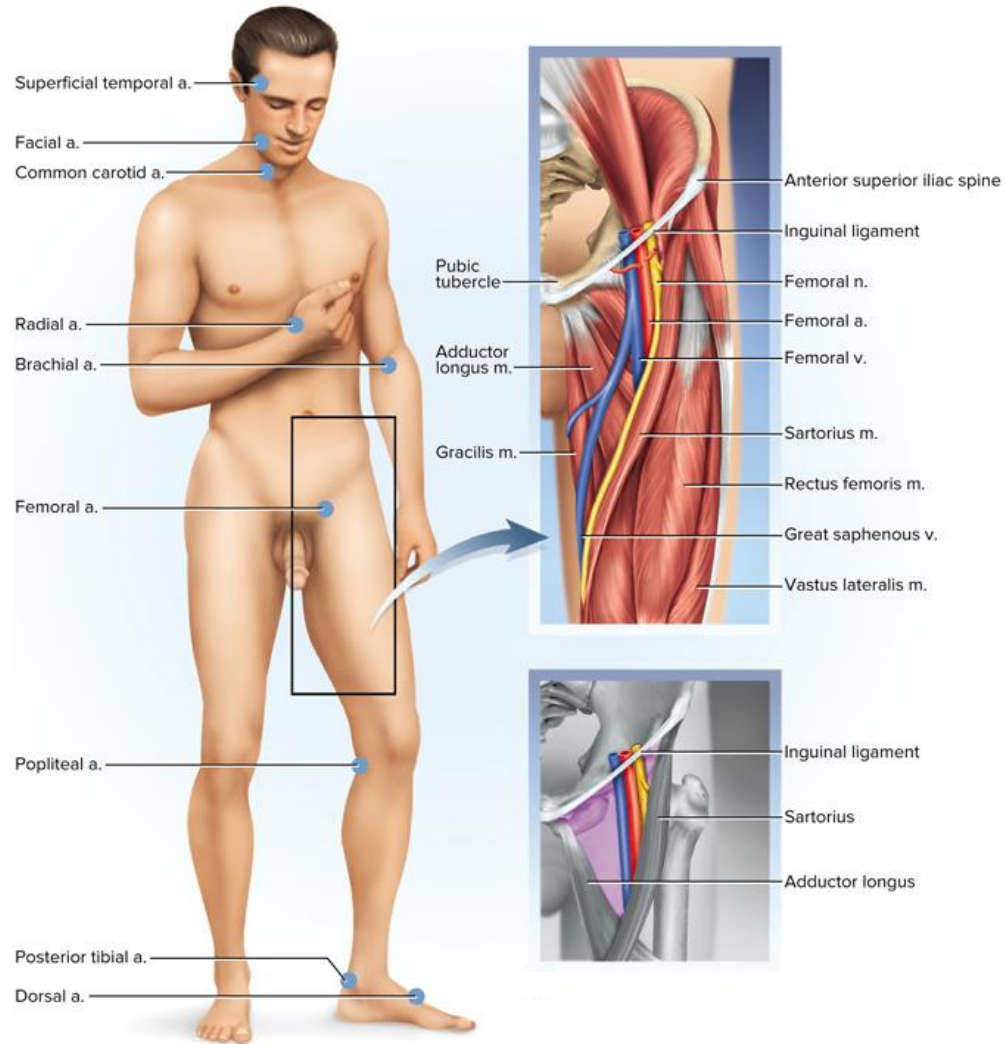
* **cardiac cycle** is a contraction (**systole**) and relaxation (**diastole**) of the heart (about .8s)

* **dicrotic notch** is a small secondary pressure increase in the beginning of diastole

* **first** and **second heart sounds** (S_1 and S_2) are often described as “lubb-dupp” and are due to closing of the valves

* **pulse pressure** - difference between systolic and diastolic pressures (about 40 mmHg)

Pulse Points



*carotid

*radial

*brachial

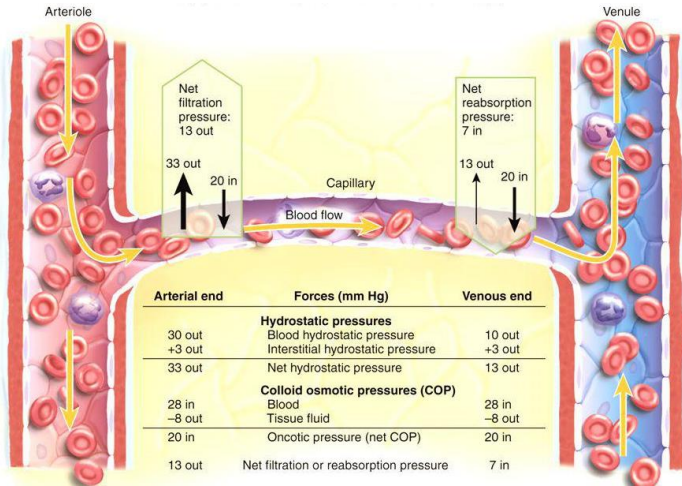
*dorsalis pedis (dorsal artery)

*posterior tibial

“Taking a Blood Pressure” Activity

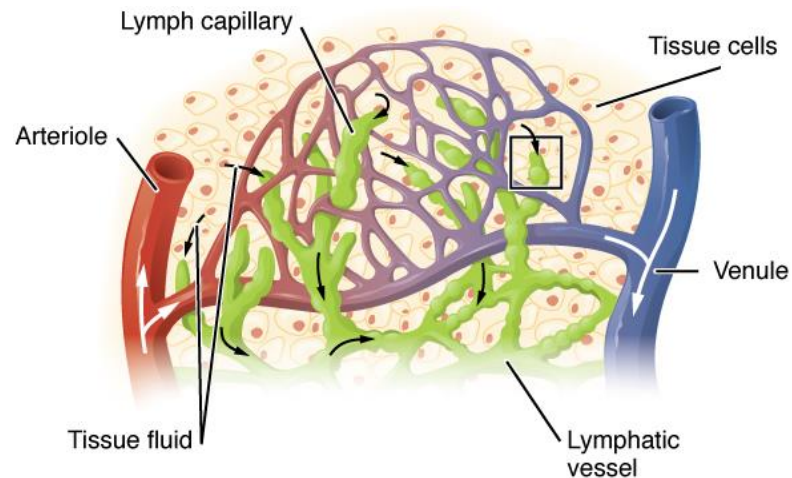
- To begin, place the cuff on your bare upper arm one inch above the bend of your elbow. Pull the end of the cuff so that it's evenly tight around your arm. You should place it tight enough so that you can only slip two fingertips under the top edge of the cuff.
- Once the cuff is on, place the disk of the stethoscope facedown under the cuff, just to the inner side of your upper arm.
- Next, place the stethoscope earpieces in your ears, with the earpieces facing forward, pointing toward the tip of your nose. Rest the gauge in the open palm of the hand of your cuffed arm so that you can clearly see it.
- Then, squeeze the pump rapidly with your opposite hand until the gauge reads 30 points above your usual systolic pressure. (Be sure to inflate the cuff rapidly). Stop squeezing. Turn the knob on the pump toward you (counterclockwise) to let the air out slowly.

Lymphatic System

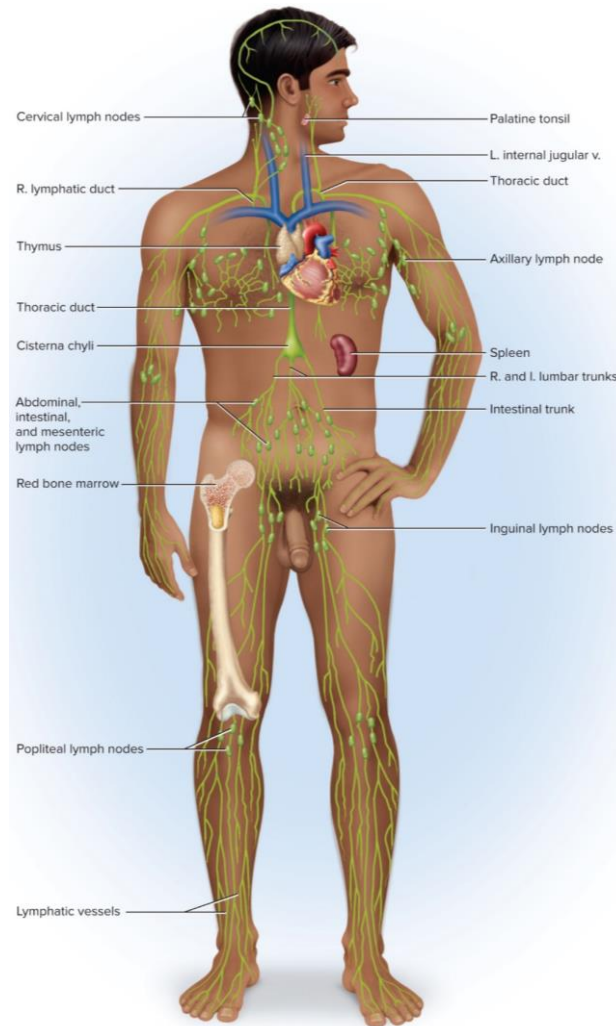


*lymphatic system has three functions

***fluid recovery** (excess fluid from interstitial space is reabsorbed - 15% - and returned to the blood)



Lymphatic System

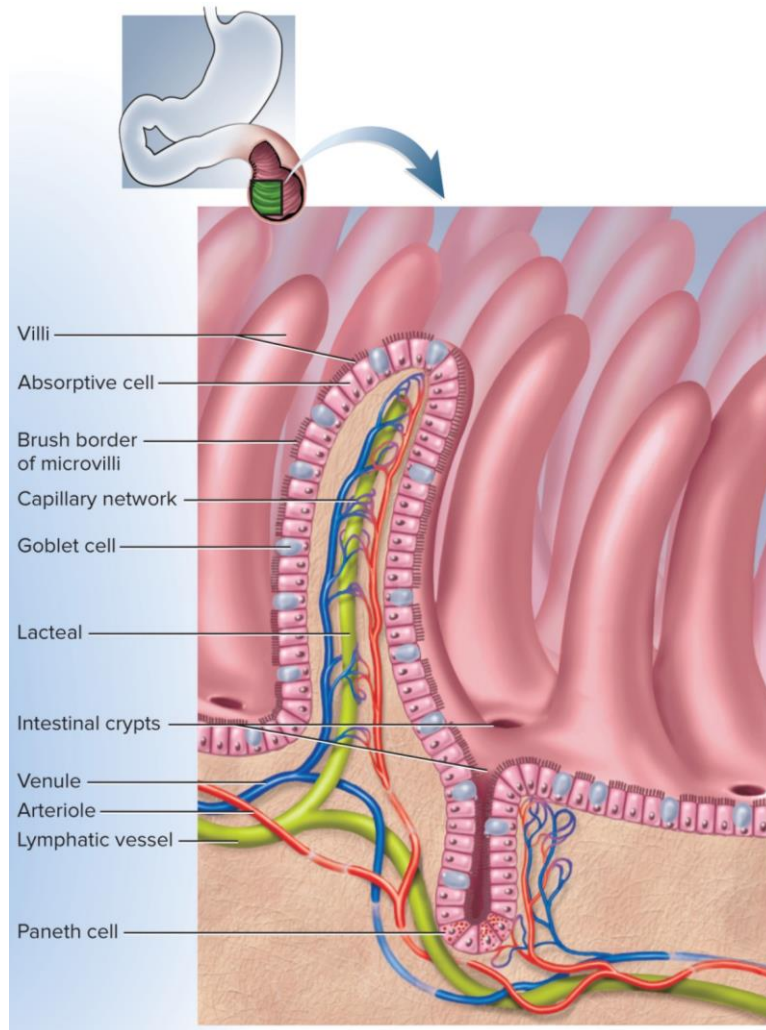


*lymphatic system has three functions

****fluid recovery*** (excess fluid from interstitial space is reabsorbed - 15% - and returned to the blood)

****immunity*** (tissue fluid is filtered through lymph nodes with immune cells on its way back to the bloodstream)

Lymphatic System



*lymphatic system has three functions

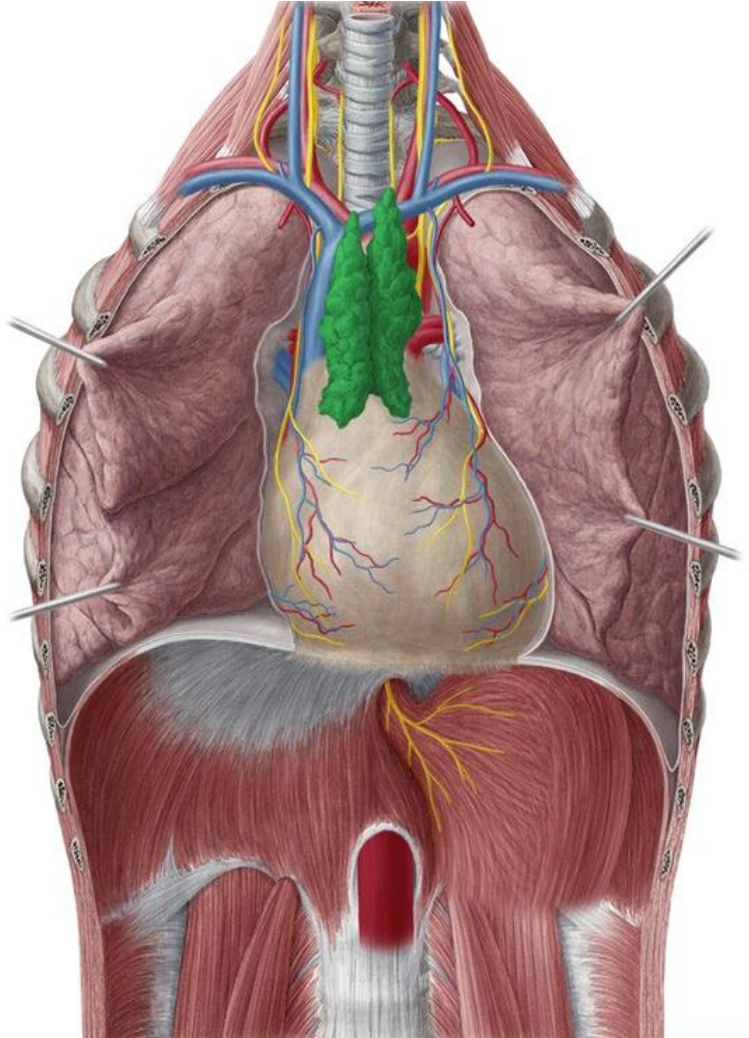
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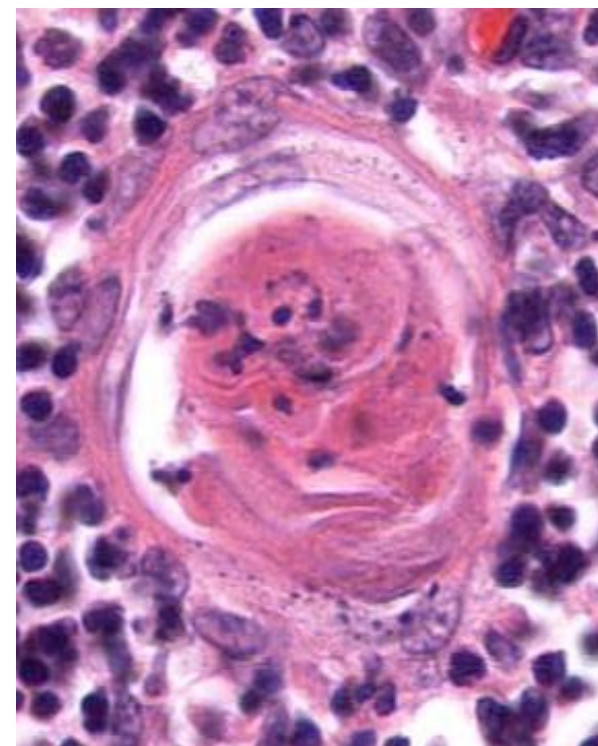
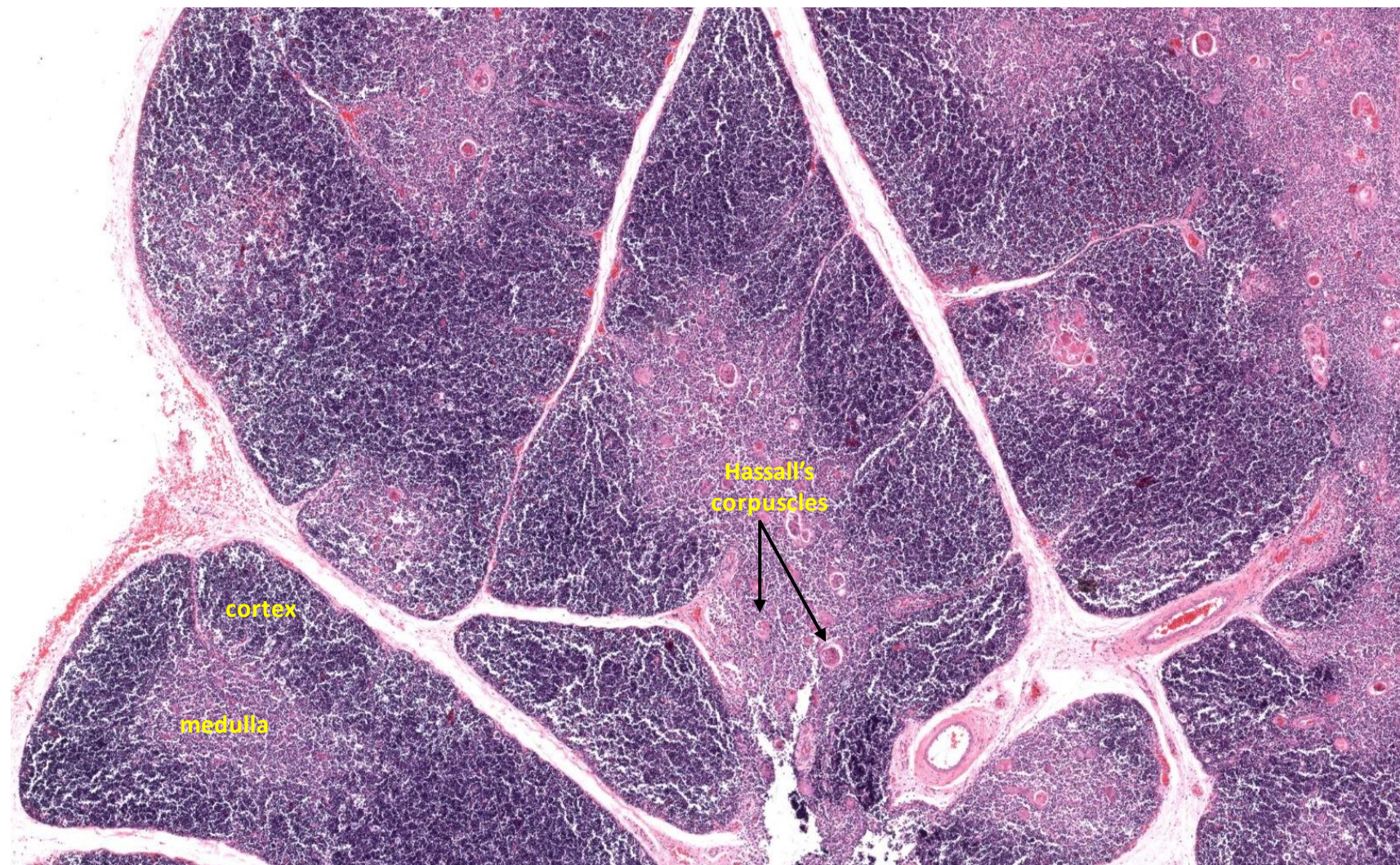
***immunity** (tissue fluid is filtered through lymph nodes with immune cells on its way back to the bloodstream)

***lipid absorption** by **lacteals** in small intestine

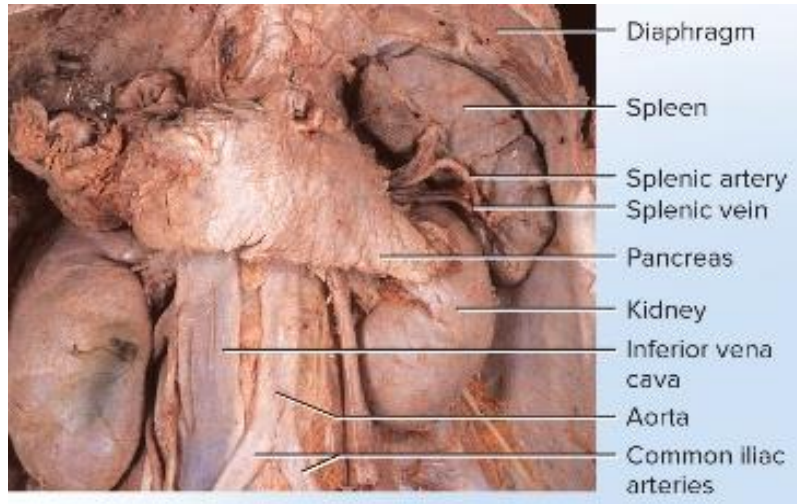
Lymphatic Organs

****thymus*** is a bilobed organ located between the sternum and aortic arch



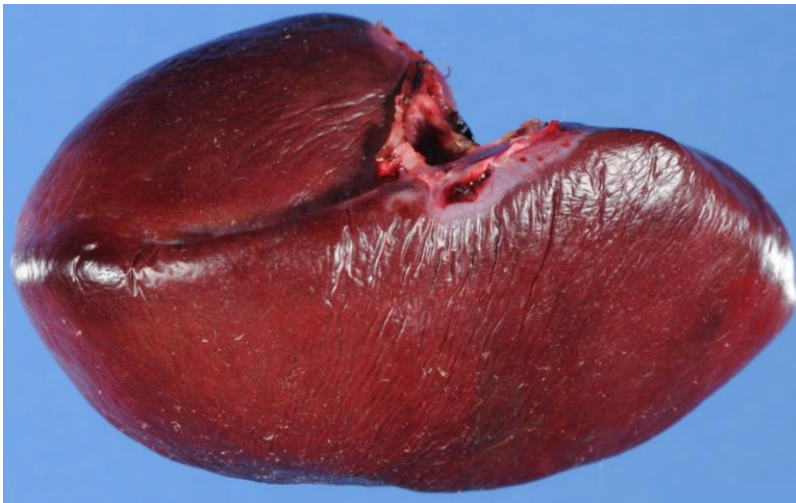


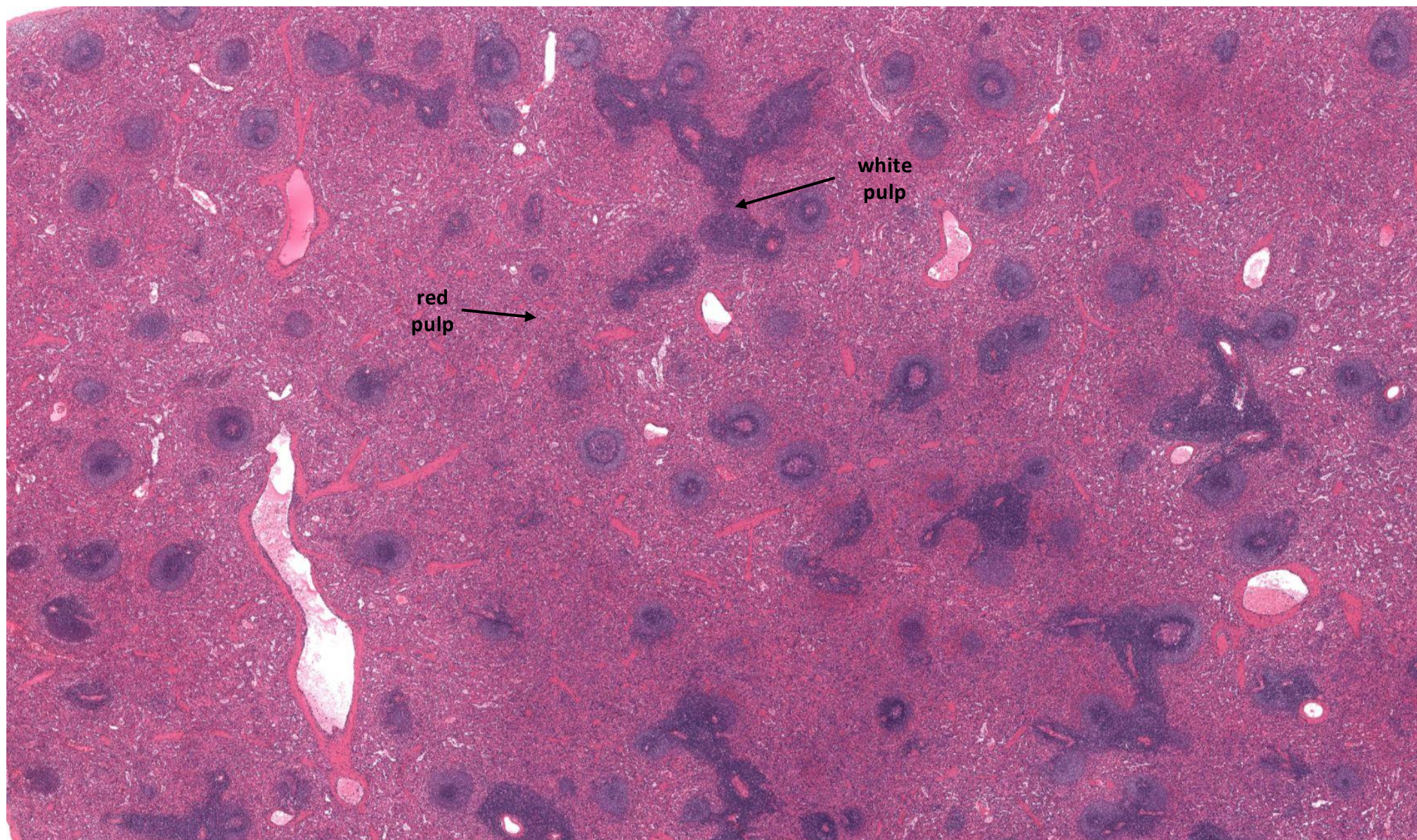
Lymphatic Organs



****thymus*** is a bilobed organ located between the sternum and aortic arch

****spleen*** is in the left hypochondriac region, inferior to the diaphragm and posterolateral to the stomach



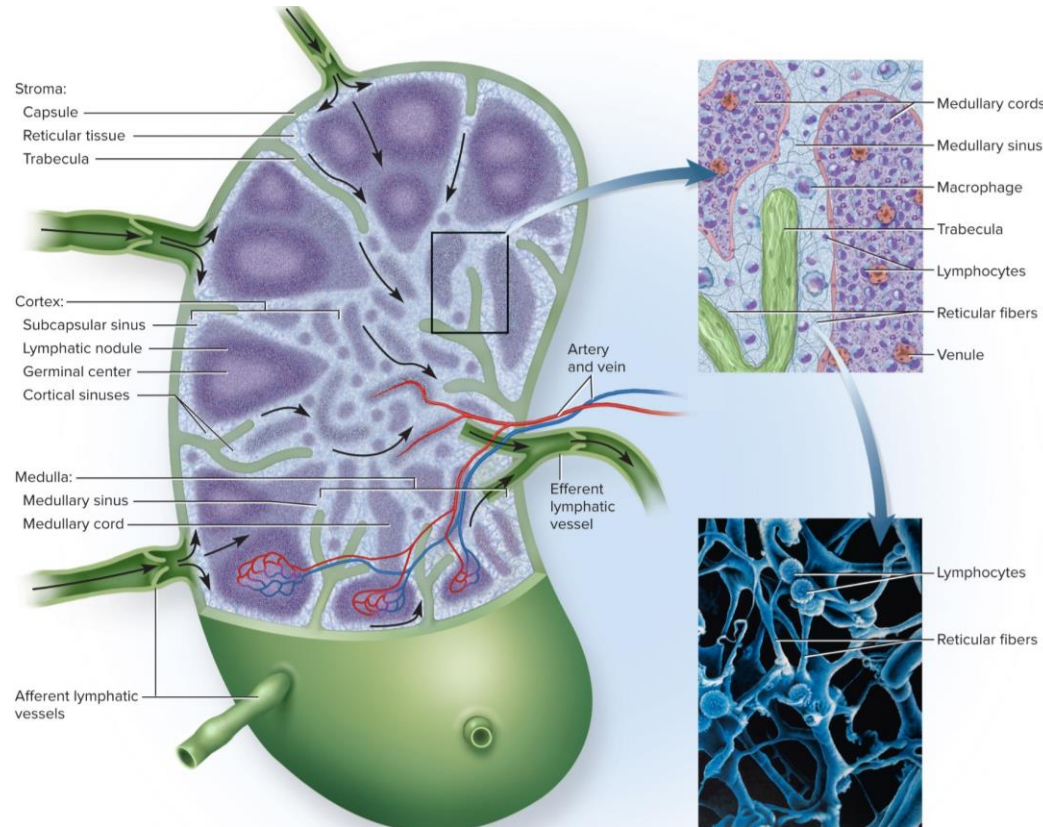


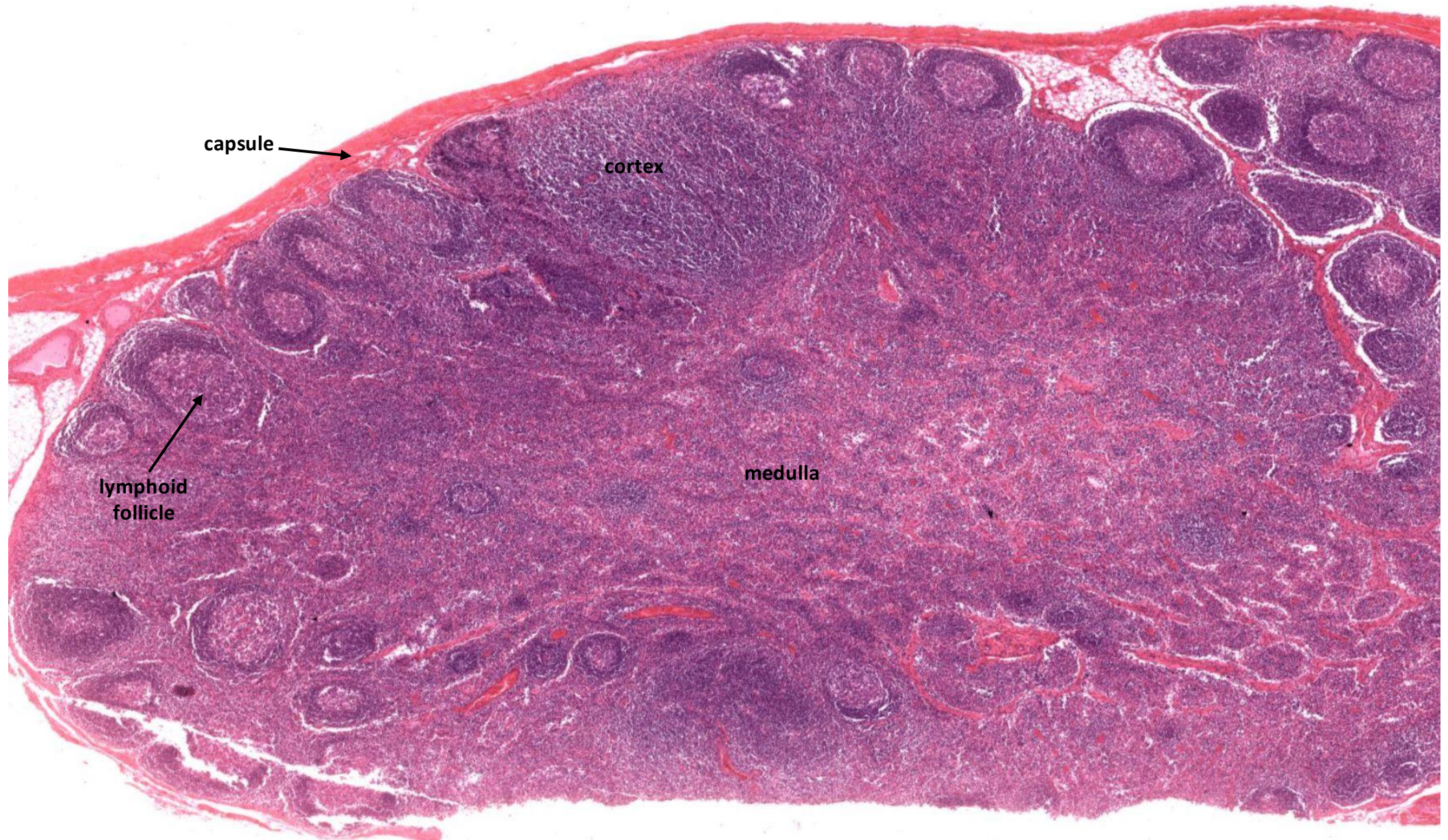
Lymphatic Organs

****thymus*** is a bilobed organ located between the sternum and aortic arch

****spleen*** is in the left hypochondriac region, inferior to the diaphragm and posterolateral to the stomach

****lymph nodes*** are the most numerous lymphatic organs (about 450 in adult)





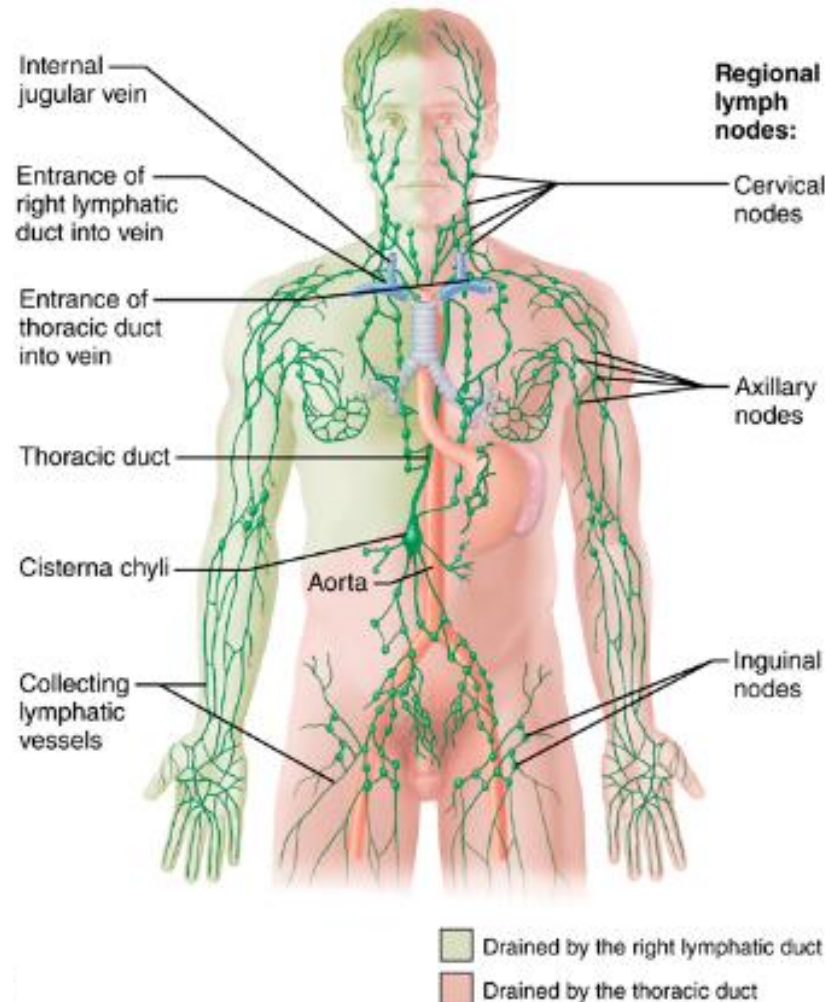
capsule

cortex

lymphoid
follicle

medulla

Lymphatic Organs



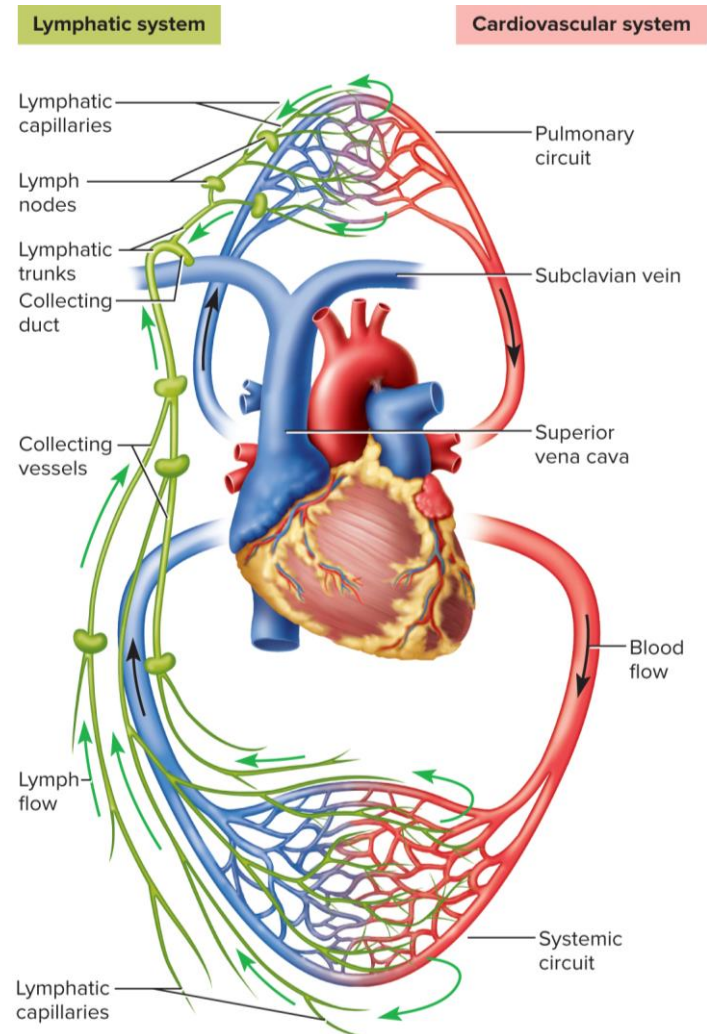
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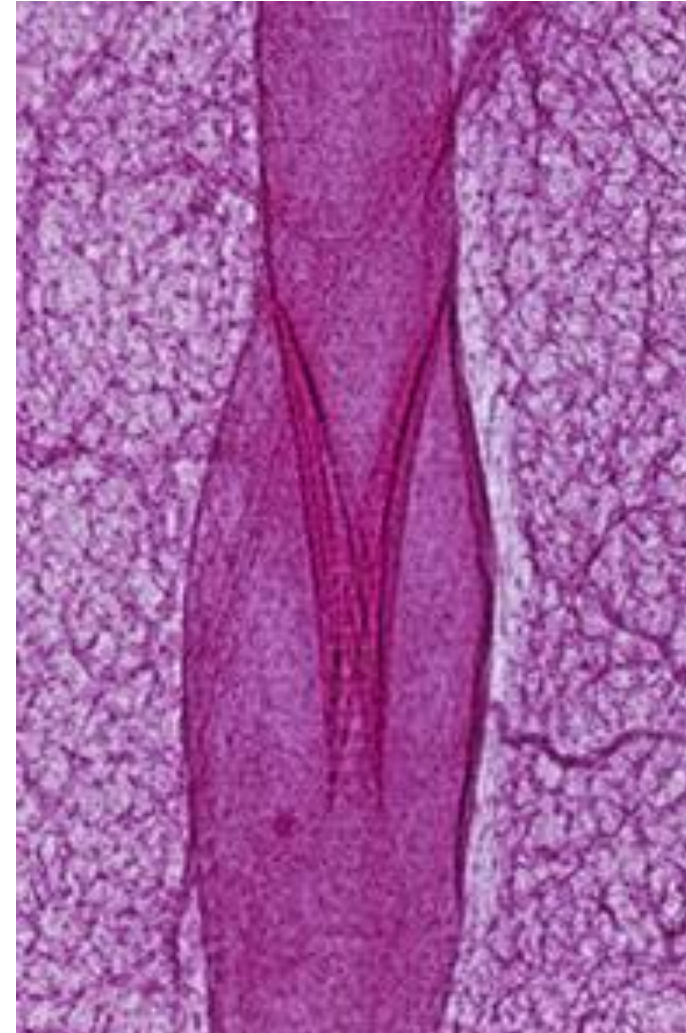
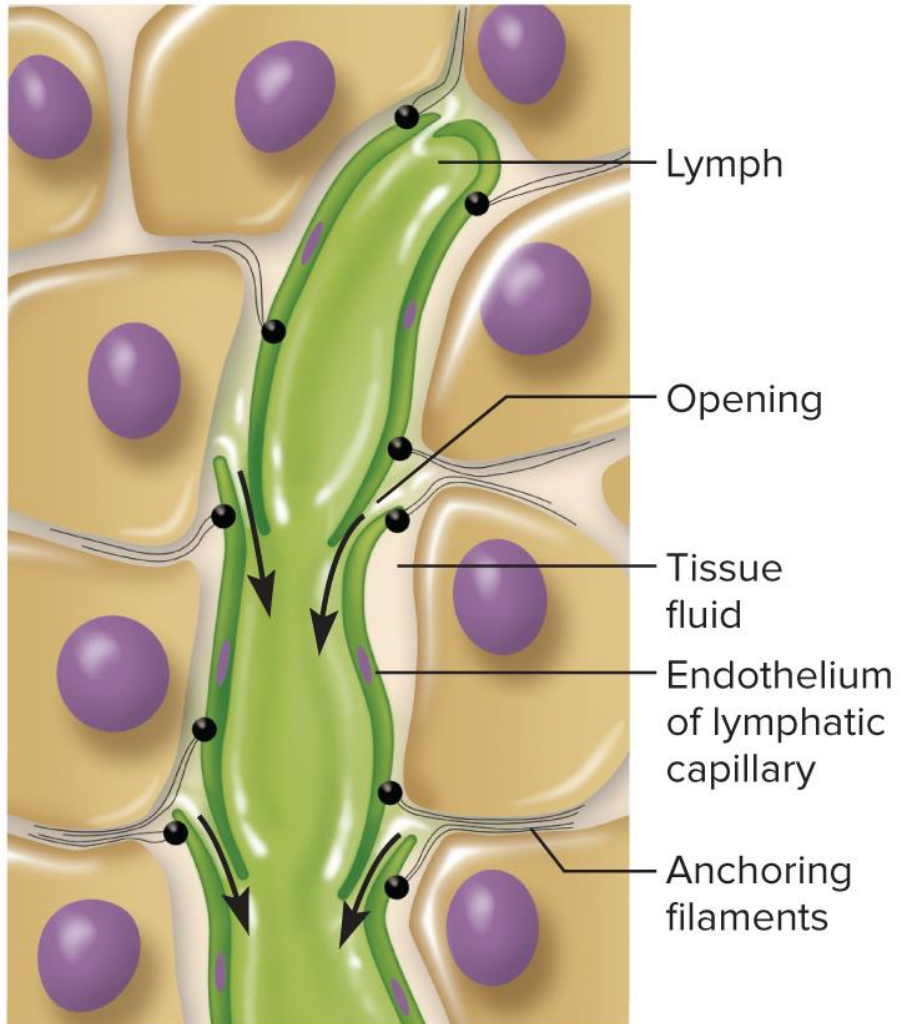
- ***cervical***
- ***axillary***
- ***intestinal***
- ***inguinal***

Lymphatic Organs

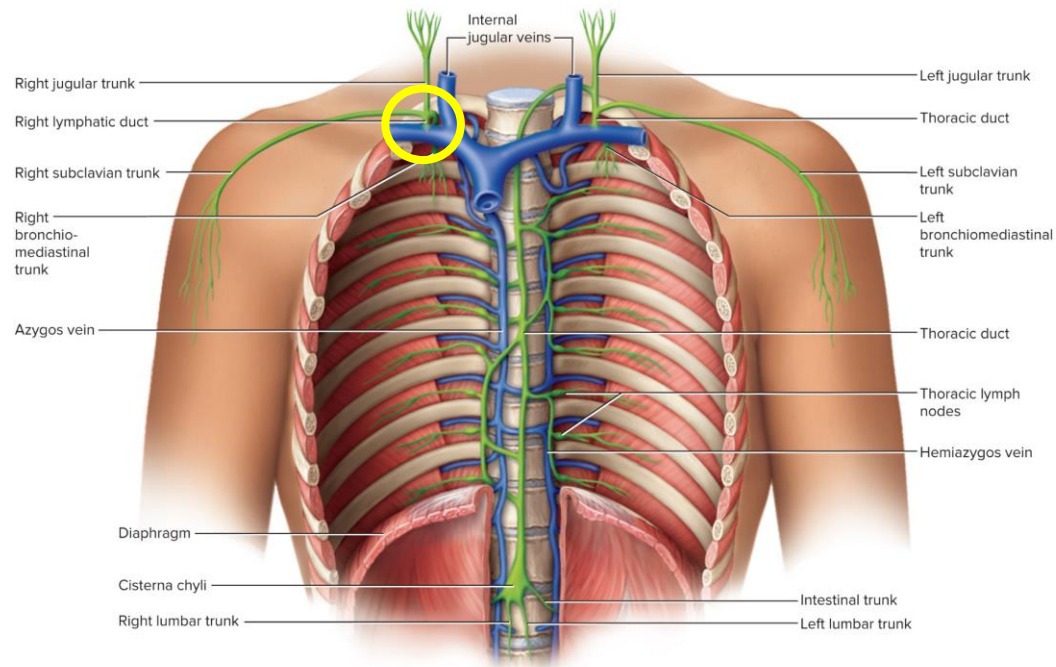


*lymph flows through a series of ***lymphatic vessels (lymphatics)***

Lymphatic Organs

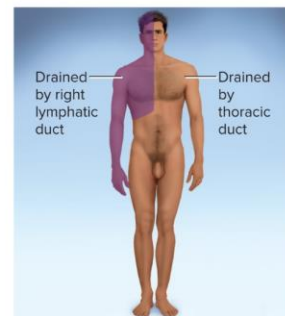
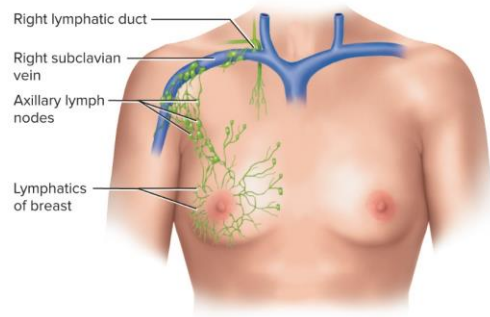


Lymphatic Organs

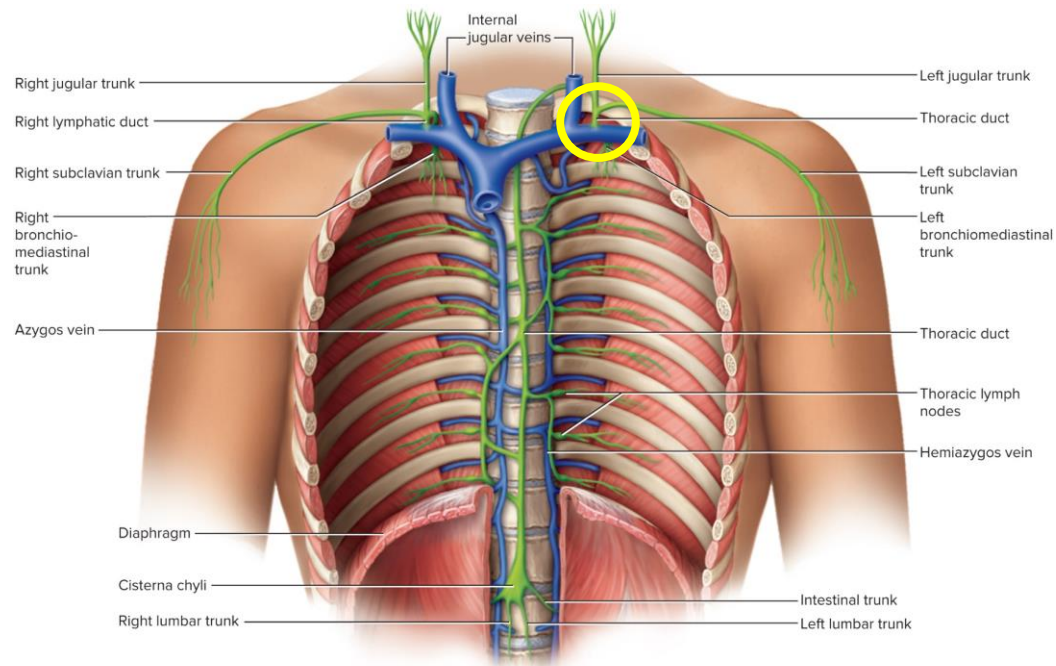


*lymph flows through a series of ***lymphatic vessels (lymphatics)***

****right lymphatic duct*** is formed by the convergence of the right jugular, subclavian, and bronchomediastinal trunks



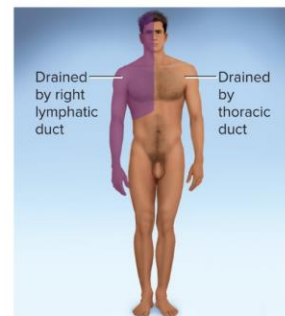
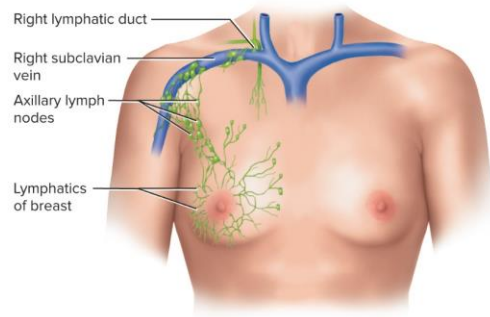
Lymphatic Organs



*lymph flows through a series of ***lymphatic vessels (lymphatics)***

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****thoracic duct*** drains from a sac called the ***cisterna chyli*** and empties into the left subclavian vein



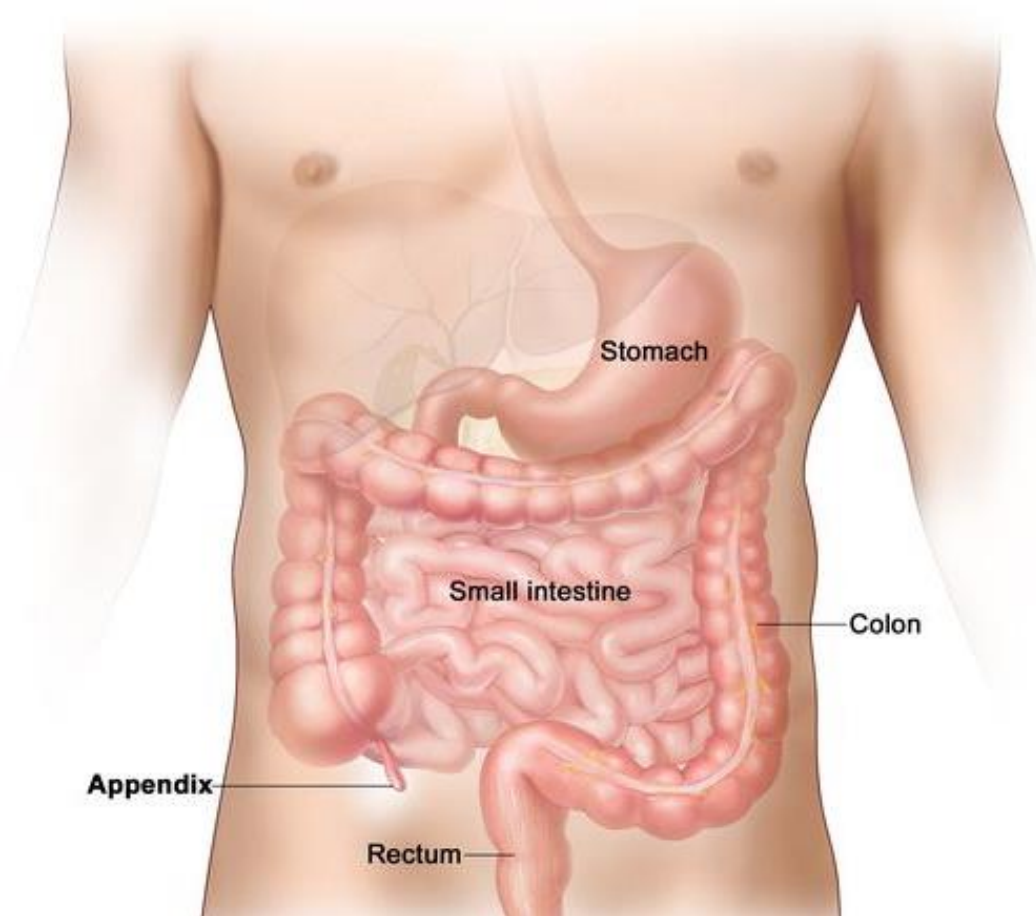
Lymphatic Organs

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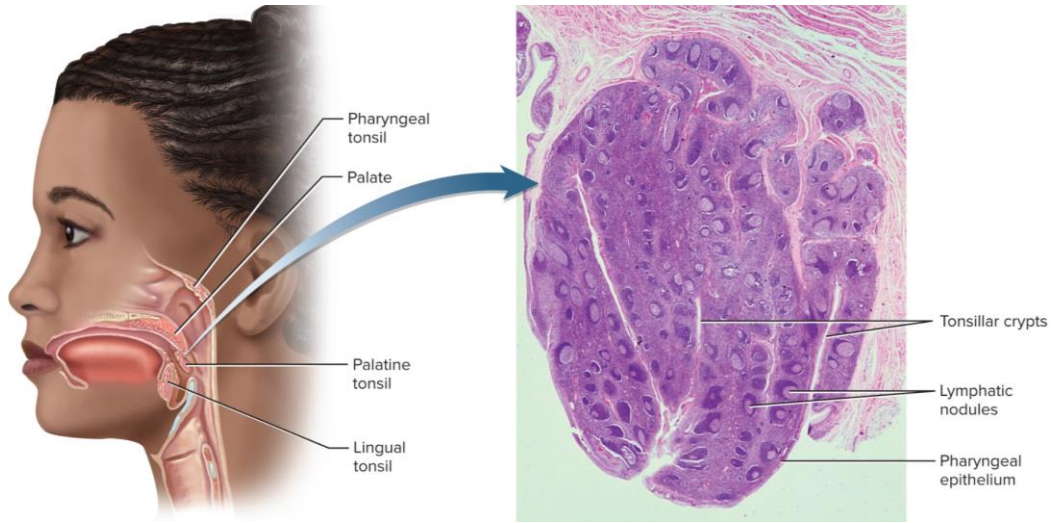
****appendix*** contains abundant ***lymphatic nodules (follicles)***

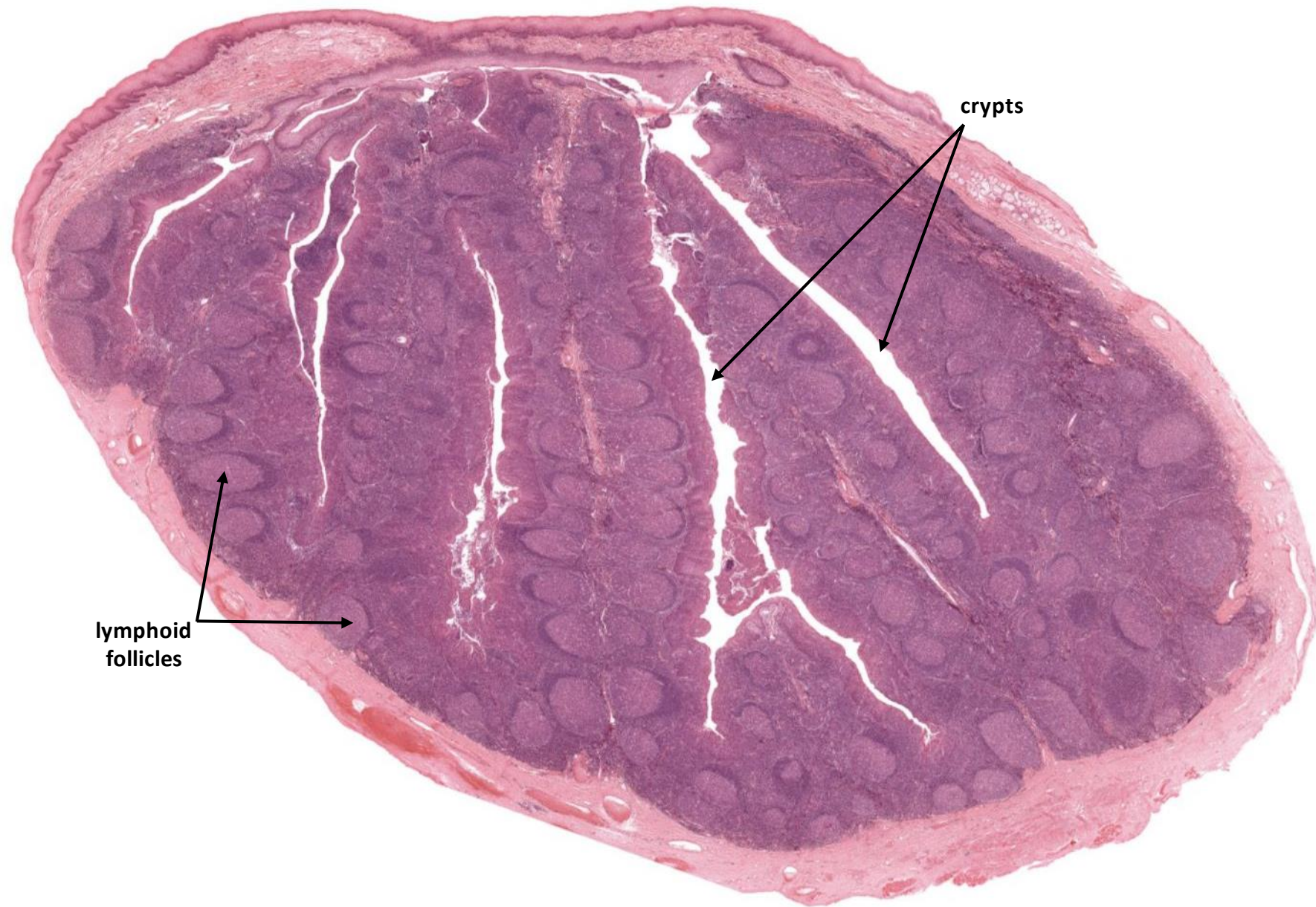


Lymphatic Organs

*three main sets of tonsils

- a single ***pharyngeal tonsil (adenoids)***
- a pair of ***palatine tonsils***
- numerous ***lingual tonsils***





crypts

lymphoid
follicles